essentially weightless backing ($\psi_{back} = 1$) or with a backing sufficiently thick as to have saturation. The saturation thickness corresponds to approximately 20% of the range of the β-particles in the backing material. Scattering can also occur from the walls of the sample holder, but this is usually less important than the backscattering from the sample backing material.

If the sample is thick, the count rate may be increased by internal backscattering, but the decrease of the count rate due to self-absorption within the sample is a greater factor. This sample self-absorption is inversely proportional to the β-ray energy and directly proportional to the thickness of the sample. For β^- from ^{32}P (1.72 MeV) sample thicknesses of 15 mg cm^{-2} show little self-absorption, while for β^- from ^{14}C (0.15 MeV), the absorption for sample thicknesses as small as 1 mg cm^{-2} is significant. The absorption factor can be determined by counting a series of samples of different thicknesses but with the same total disintegration rate, and then extrapolating that to zero sample weight.

The absorption factor ψ_{abs} is related to the absorption of the particles after they leave the sample by any covering over the sample, by the air between the sample and the detector, by the absorption in the detector window, etc. Again, this factor is usually determined by experimental comparison with a sample of known absolute disintegration rate, a *standard*.

9.11. SAMPLE PREPARATION

From the discussion of the factors that enter into the counting efficiency it is obvious that the preparation of the counting sample must be done with care and must be reproducible if several samples are to be compared. Counting of α- and β-emitters *in solution* is best achieved by means of liquid scintillation counting. Because in this technique the emitters are included in the detection system itself the efficiency is very high and reproducible.

The advantage of using *solid samples* for counting is that the samples can be made very robust and small, allowing the use of either very simple counting systems (e.g. GM-counters), or the use of commercial very efficient high-capacity (>1000 samples/hour) automatic counting systems. Alpha-emitters can only be counted efficiently if the sample is very thin so that the self-absorbtion is eliminated. For α-spectrometry the surface density should be <0.1 mg cm^{-2}. Preferably, α's should be counted by surface barrier detectors, windowless proportional counters or internal ion chambers. Counting of solid samples of β-emitters may or may not be a problem depending on the energy of the β-emission. Again, care must be taken with uniform thickness of sample, back-scattering, etc. The use of energy sensitive detectors makes possible a reliable measurement of one particular radioactive nuclide in the presence of other radiation of secondary importance.

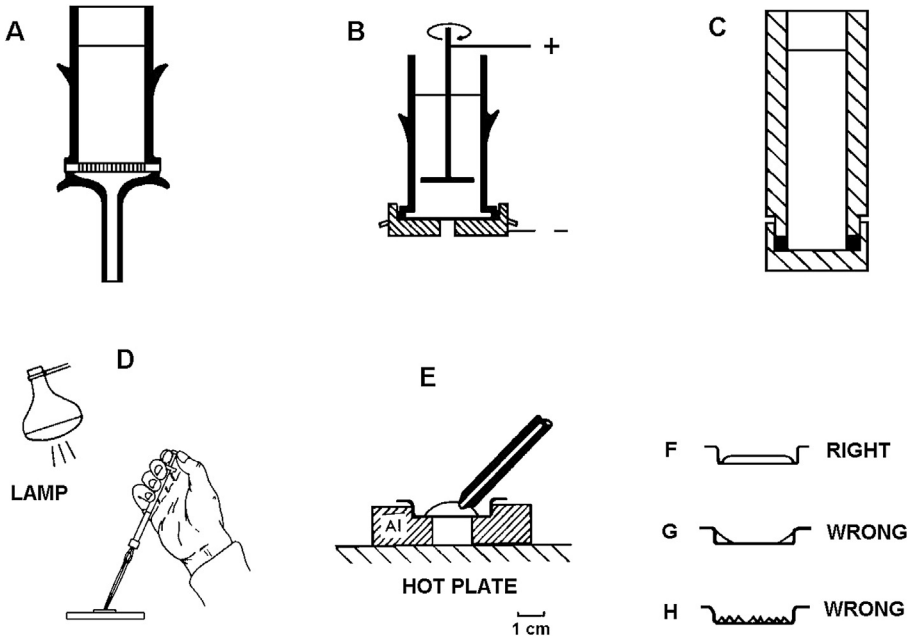


Figure 9.24 The preparation of thin, uniform samples of known surface density. A, filtering arrangement; B, electrolytic plating system; C, centrifugation vessel; D E, pipetting of a solution into an evaporation tray (glass, stainless steel, platinum), yielding a correct even deposit F or uneven samples G H. The aluminum ring E should have a temperature only slightly exceeding the boiling point of the liquid.

Solid samples can be prepared by a variety of techniques such as precipitation, evaporation, and electrolysis (Fig. 9.24). When the precipitation technique is used the radioactive material must always be precipitated for comparative counting with the same amount of carrier and all samples must have the same surface density. The precipitate is filtered on a filter plate or filter paper of known reproducible geometry A. If filtration is not feasible the precipitate may be centrifuged in special vials C, or the precipitate, after centrifugation and decanting, may be slurried with ether or alcohol, and the slurry transferred by pipette to a counting disc of fixed geometry; when the organic liquid evaporates, it hopefully leaves a uniform deposit of the precipitate slurry D, F. Problems are plentiful in obtaining a deposit of uniform thickness by evaporation of a solution. However, an arrangement such as that shown in Figure 9.24(E) has been found to be suitable; slow and even evaporation of 0.1—1.0 ml samples result in an even deposit if the amount of solid material is small. More even deposits can be obtained by electro-deposition of samples from solution B. This method can be used also for nonaqueous solutions provided that the organic solvents contain traces of water and a potential of several hundred volts per centimeter is used. To speed-up sample preparation, different

types of semi-automatic sample preparation systems are commercially available. They are particularly useful in bio-medical tracer research, where often large numbers of samples are produced.

9.12. STATISTICS OF COUNTING AND ASSOCIATED ERROR

Even if the experimental design and execution are perfect so that the determinant error is eliminated in experiments involving radioactivity there is always a random error due to the statistical nature of radioactive decay. Each radioactive atom has a certain probability of decay within any one time interval. Consequently, since this probability allows unlikely processes to occur occasionally and likely processes not to occur in any particular time interval, the number of decays may be more or less than the number in another similar interval for the same sample. It is necessary, when counting a sample, to be able to calculate the probability that the recorded count rate is within certain limits of the true (or average) count rate.

The binomial distribution law correctly expresses this probability, but it is common practice to use either the Poisson distribution or the normal Gaussian distribution functions since both approximate the first but are much simpler to use. If the average number of counts is high (above 100) the Gaussian function may be used with no appreciable error. The probability for observing a measured value of total count N is

$$P(N) = (2\pi\overline{N})^{-½}\, e^{-½\left(\overline{N}-N\right)^2/\overline{N}} \tag{9.33}$$

The standard deviation σ is given in such cases by

$$\sigma = \sqrt{N} \tag{9.34}$$

In these equations $P(N)$ is the probability of the occurrence of the value N while $\overline{N}$ is the arithmetic mean of all the measured values and N is the measured value. Figure 9.25 shows the Poisson and Gaussian distributions for $\overline{N}= 20$ counts. The standard deviation ("statistical error"), according to the theory of errors, indicates a 68% probability that the measured value is within $\pm\sigma$ of the average "true" value $\overline{N}$. For 100 measured counts the value 100 ± 10 indicates that there is a 68% probability that the "true" value will be in the interval between 90 and 110 counts. If the error limit is listed as 2σ the probability that the "true" count will be between these limits is 95.5%; for 3σ it is 99.7%. Figure 9.26 shows the relationship between K, the number of standard deviations, and the probability that the true figure lies outside the limits expressed by K. For example, at $K = 1$ (i.e. the random error is $\pm1\sigma$) the Figure indicates that the probability of the true value being outside $N \pm \sigma$ is 0.32 (or 32%). This agrees with the observation that the probability is 68%, i.e. the "true" value is within the limits of $\pm1\sigma$. Figure 9.25 can be used to establish a "rule of thumb" for rejection of unlikely data. If any measurement